

Apoorv Sharma

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EXPERIENCE

- **Amazon Web Services** Bengaluru, India
Software Developer 2 - OpenSearch July 2021 -
 - **OpenSearch containerization:** Containerized the OpenSearch process. This allows version updates to be performed in-place instead of requiring blue-green deployments which take time, duplicate infra, pose data loss risk
 - * Updates now consume constant time instead of being dependent on the data store size, decreasing from potentially 36 error prone hours to 1 hour per cluster. $O(n) \rightarrow O(1)$
 - * The newfound isolation also allows for new features to be built into the service, e.g., custom dictionaries. Cost savings for customers - upwards of 2.4M USD.
 - **AutoSync:** Programmed sync of commits GitHub and internal GitFarm for OpenSearch and its plugins. Automated new branch syncs, commit conflict resolutions, ticketing for manual intervention, building artifacts.
 - * Brought release cadence of 30 versions down from 6 months to 2 months.
 - **Disk Optimization:** Optimized the disk space of the 20 GB root volume of Amazon OpenSearch Service clusters by 10%! Generated directed acyclic graph to deduplicate dependencies and locked down the capacity allocated to system logs, tuning local retention and rotation.
 - * Projected annual cost savings: \$5.7M.
 - **Verbose logging:** Released an improvement to the company-wide default amazon logger that includes line number, file name of each logged statement. This greatly enhances the debugging experience by facilitating logic tracking.
 - **Ownership:** Assumed ownership of all projects. This includes pitching the project to SDMs and PEs, designing, implementing and deploying solutions globally. Mentored 1 developer to promotion and 2 interns to conversion.
- **Societe Generale** Bengaluru, India
Specialist Software Engineer June 2017 - May 2021
 - **Reboot as a service:** Rebootas is a scalable asynchronous service to automate execution of commands on a batch of upto a 1000 hosts all at once in parallel, eliminating manual engineer maintenance.
 - **Decom Automation:** Automated the decommission process of servers/services, including config collection, pre-checks, and workflow creation in inventory database.
 - **JSON Web Token for authentication and authorization:** Provided a much needed alternative to the company's internal authorization system which doesn't allow automation.
 - **PWStore:** A service that allows clients and stakeholders to store credentials and any other confidential data that is used in services without the need to involve developers. All the data is stored in encrypted form.
 - **Build automation:** Automated end-to-end provisioning of physical machines and VMs, covering client request-sheet validation, new IP registration, deployment file generation, and first-boot OS setup.
- **Tata Consultancy Services** Noida, India
Software Developer Intern June 2016 - July 2016
 - **Code Quality:** In charge of code review and functional testing. Debugging for E-forms in active deployment. Part of IRD and E-form design brainstorming and creation.

EDUCATION

- **Vellore Institute of Technology** Vellore, India
B.Tech. Computing Science and Engineering; CGPA: 8.4/10.0 July 2013 - May 2017

SKILLS

- **Programming:** Python, C++, Java, TypeScript, JavaScript, SQL, NodeJS
- **Cloud & DevOps:** AWS (EC2, EBS, S3, OpenSearch, Lambda, IAM, CloudFormation, CDK, CloudWatch), Docker, Jenkins, Git, CI/CD, Kafka
- **Databases:** MySQL, PostgreSQL, DynamoDB, MongoDB, NoSQL, Cassandra, GeoHashing, BKD Trees
- **Systems & Architecture:** Distributed Systems, System Design, Microservices, Event-Driven Architecture, OOPS, UNIX/Linux, 2 Phase Commit, SAGA pattern
- **Practices:** Agile, Test-Driven Development (TDD), REST, Observability (Logging, Metrics, Tracing), Estimation using HyperLogLog, Count-Min-Sketch, Bloom filters
- **Machine Learning:** Neural networks, Transformers, Regression, ML engineering, ML system design, Vectorization, RAG